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RESEARCH ARTICLE

NOMA-Enabled UAV Trajectory Optimization Using Deep Reinforcement Learning for Efficient IoT Data Collection

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ABSTRACT Non-orthogonal multiple access (NOMA) is a promising technology for future networks because it can allow many sensors to connect and share the same resources at the same time. This provides a spectrally efficient method in situations where spectrum is limited but number of sensors are enormous. In this paper, we extend NOMA to unmanned aerial vehicles (UAVs) to solve their limited flight time issue while collecting data from sensors that are distributed randomly across an open field. The objective of this study is to design an efficient UAV trajectory planning scheme that maximizes the total collected sensor data within the available flight time. To achieve this, we model the UAV trajectory optimization as a Markov Decision Process and employ a Deep Q-Network (DQN) to adaptively select UAV movements. A novel reward function is designed to simultaneously guide the UAV toward sensor locations, avoid unnecessary detours, and enhance data collection efficiency under uplink NOMA. Simulation results demonstrate that the proposed NOMA-enabled DQN framework achieves up to 93.5% average data collection, offering an improvement of approximately 19% compared to the traditional TDMA-based approach, while requiring no retraining when the sensor distribution varies.

INDEX TERMS Non-orthogonal multiple access, UAV, sensors, IoT, deep reinforcement learning, Markov decision process.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) have found applications in various domains, ranging from entertainment and photography to agriculture and surveillance. However, their potential extends far beyond these conventional applications. UAVs have found a new and promising role in the sensor network, where they can be utilized for data collection, monitoring, and communication. The integration of UAVs and sensors promises to revolutionize data collection for various sectors such as environmental monitoring, smart cities, precision agriculture, and disaster management. Sensors generate vast amounts of data that can provide valuable insights and assist in decision-making. Traditional data collection from distributed sensors poses significant logistical challenges. Fixed data collection stations, wired infrastructure, and ground-based vehicles are some common methods used to gather data from remote or inaccessible locations. However, these methods are often limited by their static nature, high operational costs, and limited adaptability to dynamic environmental conditions. With the emergence of UAVs, it has become possible to collect data from sensors dynamically without the need for much physical infrastructure. UAVs offer a unique advantage in the collection of internet of things (IoT)

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data due to their mobility, flexibility, and ability to reach remote or hard to reach areas.

However, efficient data collection from sensors by UAVs poses several challenges. One of the most important challenges is trajectory optimization, which involves determining the optimal path for the UAV to traverse while collecting data from sensors. The trajectory must consider various factors, including IoT sensor distribution, energy constraints of the UAV and characteristics of the wireless communication channel between UAV and sensors. Traditional methods include either energy harvesting using wireless signals [1], however, this may be inefficient. Furthermore, another method includes using pre-defined paths or manual planning, which are not optimal for dynamic scenarios or when dealing with a large number of randomly distributed sensors.

Reinforcement Learning (RL) is a machine learning technique in which an agent gains the ability to make successive decisions through interactions with its surroundings. RL became popular since it can manage complex tasks and dynamic decision making. In this context, Deep Reinforcement Learning (DRL) [2] can be used to learn and adapt trajectory planning strategies based on real-time feedback and environmental conditions. This approach promises to enhance the efficiency and adaptability of UAV data collection from sensors. Given that UAV trajectory planning requires sequential decision-making in a dynamic and partially unknown environment, Deep Reinforcement Learning provides an effective framework that allows the UAV to adapt its movement strategy online without the need for retraining, unlike conventional optimization-based or fixed path-planning approaches.

Building upon these motivations and existing challenges, we next discuss how UAV-enabled data collection can be further enhanced by integrating non-orthogonal multiple access (NOMA) into the trajectory design problem. The advantage of NOMA lies in the fact that it accommodates multiple users in the same time and frequency by using the power/code domain to distinguish between different users [3]. NOMA has proven to improve the spectral efficiency in scenarios with limited resources and large number of users using efficient channel estimation [4]. By applying NOMA principle to UAV-IoT scenario, we can optimize data collection in situations where there are a large number of simultaneously active sensors ready to transmit data. Therefore, this work presents a novel method to optimize UAV trajectory for data collection from sensors, utilizing DRL, Markov Decision Process, and power-domain uplink NOMA. UAV-assisted data collection faces two critical challenges in practical IoT deployments: the limited flight time of UAVs and the inefficiency of traditional orthogonal multiple access schemes in supporting massive sensor connectivity. These challenges create a fundamental tension between the need to maximize collected data and the constraints of UAV endurance. Motivated by this gap, our research explores how integrating NOMA with reinforcement learning-based trajectory optimization can significantly improve data collection efficiency, while remaining adaptable to variations in sensor deployment.

The remainder of the paper is organized as follows. Section I-A discusses existing work in this field. Section II and III discuss the proposed system model and the proposed methodology. Further, Section IV presents the simulation and results, and Section V concludes the work.

TABLE 1. Qualitative comparison between related works and proposed framework.

Work	Access Scheme	Trajectory Method	Adaptability
Conventional UAV data collection	TDMA	Predefined / heuristic	Limited
RL-based UAV trajectory (existing)	TDMA	DQN / DDPG	Retraining needed for changes
Proposed framework	**Uplink NOMA**	**DQN with adaptive reward**	**Robust, no retraining**

A. RELATED WORK

There exist multiple approaches for efficient UAV data collection from sensors. Researchers have utilized various algorithms such as those based on graph theory, optimization theory and artificial intelligence [5]. Authors in [6] also conducted a comprehensive survey on application of reinforcement learning on different UAV communication scenarios. [7] used a combination of dynamic and sequential convex programming techniques for optimizing UAV's trajectory. In [8], an initial trajectory is computed using a probabilistic Line-of-Sight (LoS) channel model before the UAV's mission, followed by online optimization. [9] presented the advantages of using neural network over table-based Q-learning for a UAV serving as a base station for two ground users. [10] presented a Deep Deterministic Policy Gradient (DDPG) algorithm for simultaneous data collection by ground and aerial vehicles using convolutional processing on an environmental map. [11] developed two techniques for trajectory optimization in accordance with state-action-reward-state-action and Q-learning to improve the efficiency of a UAV's flying route without prior knowledge of the topology of the network. [12] developed a method for minimizing the flight route and maximizing the amount of collected data by using deep Q-learning. An Ant colony optimization-based path-planning technique is presented in [13] to reduce data collection time, maximise UAV per-node capacity, and increase data collection. [14] presented a trajectory planning method based on double deep Q-network with combined experience replay in order to maximise the amount of data collected from sensors. This method takes into account various environmental constraints such as the maximum flight time of the UAV and obstacles. The agent uses Convolutional Neural Network layers to process a multi-layer map of the surroundings for deciding how to movement of the UAV. Additionally, the map is

centered on the UAV's location, which improves the agent's learning efficiency. The key methodological differences between existing studies and the proposed framework are summarized in Table 1. Although exhaustive, however, the existing work poses certain level of limitations. In some cases in spite of the advantage of using neural network over traditional Q-learning, the training time becomes very long. Also in [15], the agent, i.e., unmanned aerial vehicles (UAV) must be re-trained when any one of the environmental parameters changes. The work in [11] does not take into consideration the flight time constraint of the UAV. In [14], TDMA technique is used wherein data can be collected by only one IoT at a given time thus reducing data collection efficiency. Although several DRL methods such as DDPG, Double-DQN, and dueling architectures have been applied in UAV communication contexts, their performance is highly dependent on environment design and reward shaping, which complicates direct one-to-one comparisons. In this work, we therefore adopt TDMA as the baseline, since it is the most widely used and well-understood orthogonal multiple access scheme. This choice provides a clear and standard benchmark to demonstrate the performance improvement achieved by integrating NOMA with UAV trajectory optimization.

This work overcomes the above limitation by utilizing NOMA and DRL in the UAV-IoT network by developing a novel reward function such that the agent (UAV) travels in an optimal path maximizing data collection from sensors.

B. MOTIVATION AND CONTRIBUTION

In practical IoT deployments, large numbers of ground sensors generate continuous streams of data that must be collected within stringent time and energy constraints. UAVs offer flexibility in covering wide areas, but their limited flight endurance severely restricts the amount of data that can be collected before battery depletion. Conventional orthogonal multiple access methods, such as TDMA, further exacerbate this limitation since only one sensor can be served at a time, leading to inefficient use of spectrum and prolonged mission duration. As IoT networks continue to scale, these limitations result in substantial data loss and reduced system reliability. Motivated by this gap, we investigate how power-domain NOMA can be integrated with reinforcement learning-based UAV trajectory design to enable simultaneous data uploads, reduce redundant movement, and maximize the amount of data collected within the UAV's limited operational time. The main contributions of this paper are summarized as follows:

- This work investigates the integration of uplink NOMA with UAV-assisted data collection to address the challenge of limited UAV flight time in large-scale IoT networks.
- The UAV trajectory optimization problem is formulated as a Markov Decision Process (MDP), and a Deep Q-Network (DQN)-based learning framework is developed to maximize the collected sensor data within the available mission time.
- A novel reward function is designed to guide the UAV toward active sensors, avoid unnecessary detours, and complete data collection efficiently, thereby enhancing both spectral and energy efficiency.
- Simulation results demonstrate that the proposed NOMA-enabled DQN scheme achieves up to **93.5%** average data collection, representing an approximate **19%** improvement compared to the conventional TDMA baseline, without requiring retraining when sensor distributions vary.

It is worth noting that practical wireless systems are subject to various hardware impairments, including phase noise, I/Q imbalance, power amplifier non-linearities, and imperfect channel state information. Such impairments may degrade the achievable performance of UAV-assisted networks and remain an open challenge for future work. Recent studies have investigated the impact of hardware imperfections in related emerging paradigms such as reconfigurable holographic surfaces and intelligent metasurfaces [16], [17], [18], [19], [20]. While our present work assumes ideal hardware for analytical tractability, incorporating these impairments into UAV–NOMA design is a promising direction for future research.

II. PROPOSED SYSTEM MODEL

The proposed system model considers K stationary sensors that are assumed to be randomly distributed in an open farm field. The position of the ith IoT sensor is denoted by (x_i, y_i). Each sensor contains data denoted by D_i. The transmit power of ith IoT sensor is denoted by P_i. The UAV is initially assumed to be present at the center of the farm field and flies at a constant altitude h. Since UAVs are battery operated, their total flying time is limited which is considered to be T seconds. The total flight time T is further divided into smaller time slots of 1 second each. At any given time slot t, the position of UAV is given by $(x_u(t), y_u(t))$. The distance between the UAV and the ith IoT sensor at time slot t is given by $d_i(t)$. Assuming that the communication channel between UAV and IoT sensor is a LOS channel and the channel gain follows free path loss model, the channel gain of ith IoT sensor at time slot t is given by $h_i(t)$. Further it is assumed that a sensor can successfully transmit data only if its received signal-to-noise ratio (SNR) at the UAV exceeds a predefined threshold. This SNR based threshold effectively defines the UAV's communication range, within which reliable data exchange can occur. It is assumed that during a given time slot t, the UAV's position remain constant. The LoS channel model is assumed between the UAV and sensors, consistent with most UAV communication studies due to the typically elevated UAV position. Furthermore, A 2D trajectory model is adopted in this work, which assumes the UAV maintains a constant altitude during operation. This simplification is widely used in UAV-assisted IoT data collection studies, as it captures the essential horizontal trajectory behavior while reducing computational complexity. The framework can be

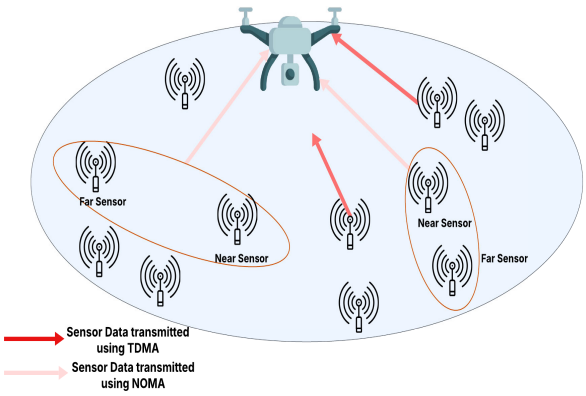


FIGURE 1. UAV-NOMA for simultaneousness sensor data collection.

TABLE 2. Summary of notations used in the system model.

Symbol	Description
K	Total number of ground sensors
T	Total UAV flight duration (time slots)
h	UAV altitude
$d_i^u(t)$	Distance between sensor i and UAV at time t
$h_i(t)$	Channel gain between sensor i and UAV at time t
P_t	Total transmit power budget for a sensor pair in a slot
$P_n(t)$	Power allocated to the near sensor at time t
$P_f(t)$	Power allocated to the far sensor at time t
σ^2	Noise power ($\sigma^2 = N_0 B$)
N_0	Noise power spectral density
B	System bandwidth
$\text{SNR}_n(t)$	Signal-to-noise ratio of the near sensor at time t
$\text{SINR}_f(t)$	Signal-to-interference-plus-noise ratio of the far sensor at time t
$R(t)$	Achievable sum rate of the selected NOMA pair at time t
D_i	Total data units to be uploaded by sensor i

extended to a full 3D trajectory optimization scenario [21], [22] but is out of scope of current work.

Figure 1 depicts NOMA employed at the UAV for pairing two sensors for simultaneous data collection. The two sensors are chosen such that one of them has a higher channel gain as compared to the other. It is evident that the sensor which is located near the UAV will have a higher channel gain whereas the sensor which is located far away from the UAV will have a lower channel gain. Therefore, at a given time slot t, a near and a far IoT sensor is searched for pairing which transmit their data simultaneously to the UAV. This makes the data collection spectrum efficient as compared to traditional multiple access techniques such as TDMA where only one sensor can transmit data at any given time. It must be noted that all sensors are assumed to be wireless and energy-constrained devices. Although detailed energy modeling is not the focus of this study, the proposed NOMA-based scheduling and optimized UAV trajectory indirectly contribute to energy efficiency by minimizing total transmission time and UAV flight duration

In each time slot, the UAV schedules exactly one NOMA group consisting of two sensors. Specifically, one sensor with relatively stronger channel gain (near user) and one with weaker channel gain (far user) are selected to transmit simultaneously. This design ensures that the UAV applies SIC to only two signals at a time, which is consistent with practical uplink NOMA implementations. Scheduling multiple NOMA groups per slot is possible in principle but would substantially increase decoding complexity. Let the near and far sensor of a NOMA pair at time slot t be denoted by n and f, respectively, and their channel gains as $h_n(t)$ and $h_f(t)$ respectively. Let the transmit powers of near and far sensor be represented as $P_n(t)$ and $P_f(t)$, respectively and their distance from UAV as $d_n^u(t)$ and $d_f^u(t)$, respectively. The relation between the allocated power and distance is inversely proportional. Therefore, if the total transmission power of UAV is P_t, then the individual transmit powers can be given by

$$P_n(t) = \frac{d_n^u(t).P_t}{d_n^u(t) + d_f^u(t)}, P_f(t) = \frac{d_f^u(t).P_t}{d_n^u(t) + d_f^u(t)} \quad (1)$$

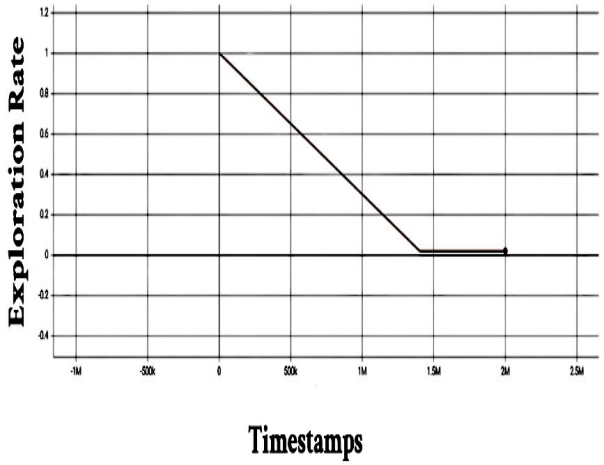
Since power-domain uplink NOMA is considered, superposition coding (SC) will be performed wherein the signals from both the sensors are combined together. Let the signals transmitted by the near and far sensors of the NOMA pair be denoted as $x_n(t)$ and $x_f(t)$, respectively. The superimposed signal received by the UAV is given as $y(t) = \sqrt{P_n(t)}h_n(t)x_n(t) + \sqrt{P_f(t)}h_f(t)x_f(t) + n(t)$, where $n(t)$ is Gaussian noise. Further, the UAV performs successive interference cancellation (SIC) on the received signal, wherein near sensor first decodes the signal from the far sensor, then it subtracts this decoded signal from the received signal, after which the signal is decoded. Therefore, the near sensor does not experience interference from the far sensor whereas the far sensor experiences interference from the near sensor. Hence, signal-to-interference-and-noise-ratio (SINR) of both the sensors can be given as

$$S_n(t) = \frac{P_n(t)h_n^2(t)^2}{n(t)}, S_f(t) = \frac{P_f(t)h_f^2(t)^2}{P_n(t)h_n^2(t)^2 + n(t)} \quad (2)$$

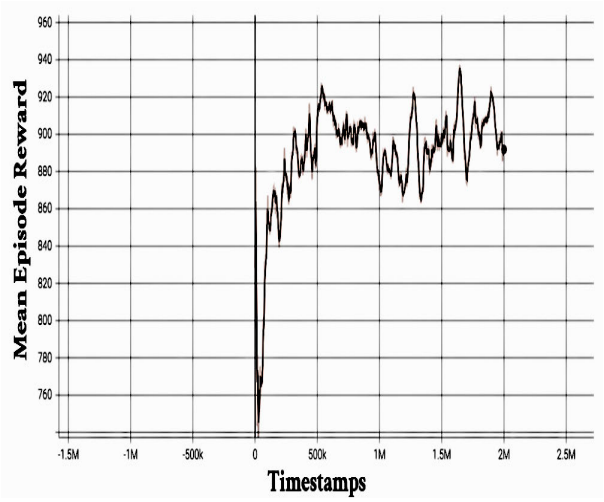
Furthermore, the data rate for near and far sensor can be given as $R_n(t) = B \cdot \log_2(1 + S_n(t))$ and $R_f(t) = B \cdot \log_2(1 + S_f(t))$, respectively, where B represents bandwidth. Since each time slot t is of 1 second, data rate in time slot t is equal to the amount of data collected in that time slot. The total amount of data collected in a particular time slot is obtained as $R(t) = R_n(t) + R_f(t)$. The total amount of data collected by the UAV throughout its flying time $\mathcal{T}$ can be calculated as

$$C = \sum_{t=1}^{\mathcal{T}} R(t), \quad (3)$$

where C is the total amount of data collected. The aim of this work is to maximize C, i.e, to collect the maximum possible data from the sensors within the given flying time $\mathcal{T}$. Hence this is an optimization problem which can be solved using MDP and DRL. The Objective function can be expressed as $\max_\pi \sum_{t=1}^{T} R_t$, where R_t denotes the instantaneous data collected at time slot t under the UAV policy π. The policy π



(a)



(b)

FIGURE 2. **(a) Decay of exploration rate during training process, and (b) Mean episode reward during training process.**

represents the mapping from observed environmental states to UAV actions, with the goal of maximizing the total data collected before the UAV exhausts its flight duration.

It must be noted that although other non-orthogonal schemes such as Sparse Code Multiple Access (SCMA) and Pattern Division Multiple Access (PDMA) have been explored in the literature, their application in UAV-assisted data collection is limited by the need for codebook/pattern design and higher signaling overhead. In contrast, NOMA leverages simple power-domain multiplexing and SIC at the UAV, making it more flexible for highly dynamic UAV–sensor networks where the set of active users changes frequently. Therefore, NOMA is adopted in this work as a practical and efficient solution.

III. PROPOSED METHODOLOGY

A. MARKOV DECISION PROCESS (MDP)

Let's denote the MDP by a tuple (S,A,R,γ,$\mathcal{T}$). Here, S represents the state space, A represents the action space, R represents the reward function, γ represents the discount factor and $\mathcal{T}$ represents the state transition probability. MDP signifies the total amount of data collected by the UAV given in (3).

- The state space is the set of all possible states. At any point of time t, the state is denoted by $s(t)$ and it consists of the position of the UAV, positions of all sensors, the amount of data left to be collected from each IoT sensor and the remaining flying time of the UAV.
- The action space denotes the set of all possible actions that the UAV can take at any point of time in any state given as $A = \{north, east, south, west\}$.
- The elements of the reward function are given below:
  - Positive reward for staying within the farm field boundary and negative reward for going outside.
  - Positive reward for moving towards nearest sensor and negative reward for moving away from it.
  - Positive reward for approaching near an IoT sensor.
  - Positive reward for collecting all the data from an IoT sensor.
- The discount factor lies between 0 and 1 (both inclusive), and balances the importance of long term and short term rewards. It prioritizes immediate rewards over future rewards in decision-making process.
- The state transition probability $\mathcal{T}$ represents the probability of the agent (UAV) taking an action "a'' in a state "s'' and moving to another state "s'''.

The designed reward function directly reflects the objective defined in the previous section. It integrates multiple components to guide the UAV's learning process effectively:

- Data rate-based reward: Provides a positive reward proportional to the achievable transmission rate between the UAV and sensors, encouraging the UAV to move toward sensors with higher throughput and thus maximize total collected data.
- Directional movement reward: Incentivizes efficient trajectory planning by rewarding movements toward unvisited or high-priority sensors.
- Penalty terms: Apply negative rewards for revisiting previously served sensors or exiting the sensing area, discouraging inefficient behavior.

By combining these elements, the reward function ensures that the UAV's trajectory optimization is aligned with the overall objective of maximizing data collection efficiency while minimizing redundant motion.

B. DEEP Q-LEARNING

For solving the above mentioned MDP, we use DQN which approximates the Q-value of each state-action pair which is given by the Bellman equation as

$$Q^{\pi}(s(t), a(t)) = E_{\pi}\left[\sum_{k=t}^{\mathcal{T}} \gamma^{k-t} R(s(k), a(k))\right] \quad (4)$$

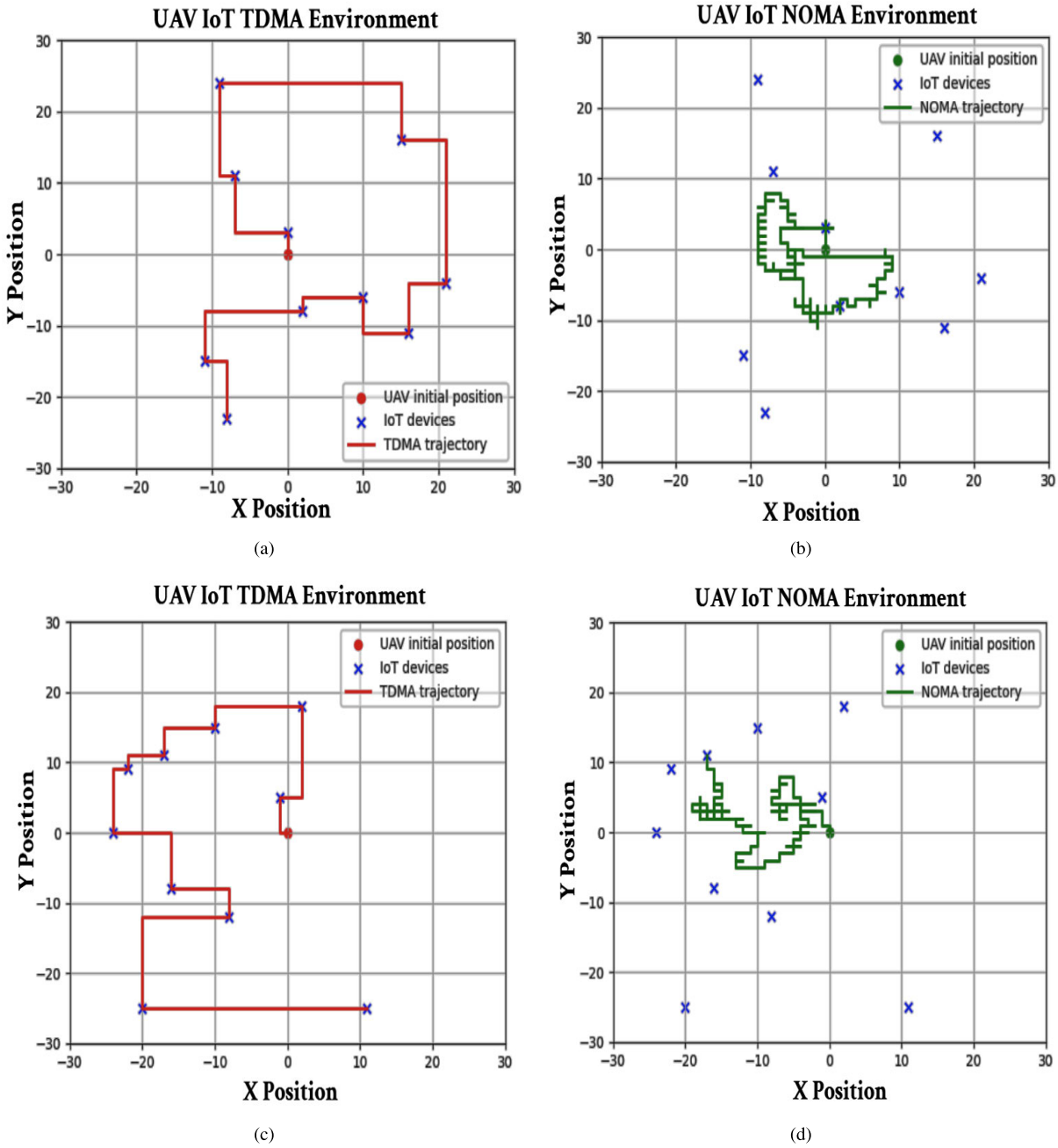


FIGURE 3. Comparison of UAV trajectories under (a) TDMA (b) NOMA (c) TDMA and (d) NOMA. In addition to the trajectory plots, the proposed NOMA approach achieves an average of 93.5% data collection in 460 steps, whereas TDMA achieves only 74.5% in 488 steps (averaged over 1000 random sensor deployments).

The above Q-value denotes the expected cumulative reward of an agent for taking action $a(t)$ in state $s(t)$ following a control policy π. The agent's goal is to learn a control policy π such that the reward it receives is maximized. For this, a Q-table of size $|S| * |A|$ is created, where $|S|$ is the total number of possible state spaces and $|A|$ is the number of possible actions. Initially, all the cells in this Q-table are filled with zeros. In normal Q-learning, the values in these cells are updated over multiple iterations as the agent explores the environment. Let s, a, s', and a' represent $s(t)$, $a(t)$, $s(t+1)$, and $a(t+1)$, respectively. Then the Q-values are updated using

$$\text{New } Q^{\pi}(s,a) = Q^{\pi}(s,a) + \psi \cdot (R(s,a) - Q^{\pi}(s,a) + \gamma \cdot max\, Q'^{\pi}(s',a')) \quad (5)$$

where $Q^{\pi}(s,a)$ represents the current Q-value for taking action a in state s. $max\, Q'^{\pi}(s',a')$ is the maximum expected reward for new state s' and for all possible actions in that state,

TABLE 3. Summary of reward function design.

Condition / Event	Reward Type	Value (Used)
UAV remains inside field boundary	Positive	$+1$
UAV exits field boundary	Negative	-5
UAV moves toward nearest sensor (directional alignment)	Positive	$+0.5 \cdot \cos\theta$
UAV moves away from nearest sensor	Negative	$-0.5 \cdot \cos\theta$
UAV reaches within proximity threshold of a sensor	Positive	$+2$
Successful data upload (sensor buffer emptied)	Positive	$+5$
Revisiting already emptied sensor	Negative	-2
Step penalty (per time slot, to encourage efficiency)	Negative	-0.1

TABLE 4. Simulation parameters.

Parameter	Value / Description
Simulation area	50 m $\times$ 50 m grid (farm field)
Number of IoT sensors (K)	10 (randomly distributed)
Initial UAV position	Center of grid $(0, 0)$
UAV altitude (h)	10 m (constant)
Total UAV flight time	500 s
Initial data per sensor	100 units
Training timesteps	2 million (1.4M for exploration, 0.6M for exploitation)
Exploration rate (ϵ)	Starts at 1, decays to 0.02 over 1.4M timesteps
Learning rate (α)	1×10^{-4}
Episode length	500 timesteps
Total episodes	4000
Evaluation scenarios	1000 random sensor deployments

ψ represents the learning rate which varies between 0 and 1 and determines the extent of how much new information overrides old information. (5) reflects the two primary KPIs optimized in this work, namely the total amount of sensor data collected and the number of UAV movement steps required, which together quantify the efficiency and effectiveness of the trajectory design.

1) STATE REPRESENTATION

At each time step, the state is represented by a feature vector that concatenates (i) the UAV's current grid position, (ii) the remaining data units at each sensor, and (iii) the instantaneous UAV–sensor distances. For the case of $K = 10$ sensors, the resulting state dimension is $1+10+10 = 21$, which is directly fed into the DQN. This design ensures that both spatial and task-related information are available for decision-making.

Since the state space for the proposed problem is large, storing the Q-values in a table is not an efficient method. Hence, we cannot use normal Q-learning to solve the proposed problem statement. However, Deep Q-Networks (DQNs) makes it feasible to reliably train massive neural networks to estimate the Q-value rather than computing and storing it in a table [23]. DQN uses experience replay to store and randomly sample experience to break correlation between consecutive experiences. It also utilizes a separate target network with delayed updates to stabilize training. Further, epsilon greedy strategy is used for balancing exploration-exploitation tradeoff so that the agent (UAV) manages to sufficiently explore the environment first and then later exploit it. This ensures that it is not stuck in local optimal reward and rather aims for global optimal reward.

The primary computational cost of the proposed framework arises during the offline training phase of the DQN. Once trained, the UAV only performs a forward pass through the neural network to select actions, which involves a small number of matrix multiplications and can be executed in real time. In contrast to optimization-based methods that require solving a new trajectory problem whenever the environment changes, this approach offers minimal online complexity with improved adaptability.

IV. SIMULATION AND RESULTS

For simulation, a 2D grid of size of $50m \times 50\ m$ is used which represents the farm field. The UAV is assumed to be initially present at the center of the grid, i.e., at (0, 0). Altitude of UAV is considered as $h = 10\ m$. The total flying time of UAV is equal to 500 seconds. 10 sensors, i.e., K=10, is considered that are randomly distributed throughout the grid. The initial data of each sensor is assumed to be 100 units. The simulation parameters are given in Table 4. The IoT sensors are assumed to be independently and uniformly distributed over the entire sensing area, ensuring no spatial bias toward any specific region and enabling a fair evaluation of the proposed UAV trajectory optimization strategy.

A custom environment is generated using OpenAI Gym. This environment has an observation space (also called state space), action space and episode length. Each episode is an entire simulation run of the UAV from its initial point till its flying time becomes zero. Therefore, episode length is equal to the flying time of the UAV. At every step of the UAV, its reward and the amount of the data that it collects through NOMA pairing is calculated. It is required that the UAV always travel towards its nearest sensor, till it either reaches that sensor, or collects all its data. It is also important to ensure that the UAV stays within the grid limits. For this, an appropriate reward function is designed wherein the dot product of the direction vector of the nearest sensor from the current position of the UAV with the direction vector of the action to be taken is calculated. A positive dot product indicates that the action to be taken by the UAV will take it towards the nearest sensor, therefore, a positive reward is assigned. A negative dot product indicates that the action to be taken by the UAV will take it away from the nearest sensor, therefore a negative reward is assigned. Moreover, to ensure that the UAV stays within the grid limits, a positive reward is awarded for movements within the grid and a negative reward is assigned for any action that takes it out of the grid. Apart from this, a set of visited sensors is maintained to make sure that the UAV does not visit same sensor twice.

Further, DQN is used to train the agent in the defined custom environment. The agent is trained for 2 million

timesteps out of which 1.4 million timesteps are for exploration and the remaining for exploitation. This is in line with the epsilon greedy strategy that has been discussed in section III. Initially, at the start of training, the exploration rate is set to 1 which then eventually decays to 0.02 over 1.4 million timesteps. This is evident from Figure 2(a). The learning rate is set as $1e-4$. The hyperparameters of the DQN training process, including the learning rate, decay rate, and replay memory size, were selected through empirical tuning to ensure convergence stability and efficient learning. The learning rate was chosen to maintain a balance between convergence speed and oscillation avoidance, while the decay rate was set to gradually shift from exploration to exploitation as training progressed. The replay memory size was kept sufficiently large to provide diverse experience sampling without exceeding computational limits. These values are consistent with those commonly reported in reinforcement learning-based UAV trajectory optimization literature.

Since each episode has length of 500 timesteps, it implies that the agent (UAV) is trained for a total of 4000 episodes. During training, the agent is evaluated after every four episodes and its average reward is calculated which is known as mean episode reward. The plot of mean episode reward during the entire training process is given Figure 2(b) and it is evident that the mean episode reward increases as the number of timesteps increases which implies that the model is learning correctly. After around $500k$ timesteps, it starts to converge. In the initial stages of the training, the agent wastes resources by exploring the grid randomly without optimizing much on the data collection. But eventually, as the training progresses, it starts exploiting the environment and thereby optimizing the data collection.

To evaluate the performance of the UAV, we test it over 1000 randomly generated scenarios. The average data collection over all these scenarios is 93.5% which is nearly 19% higher than that collected in [24]. The trajectories of UAV in two randomly generated scenario is plotted in Figure 3. The trajectory is shown using two techniques, namely, TDMA (Figure 3(a), (c)) and NOMA (Figure 3(b), (d)). It can be observed that when TDMA is used the UAV has to travel through all the sensors to collect the data. This is because at a given time UAV can only collect data from the nearest sensor while using TDMA. However, when NOMA is used at the UAV, it does not visit all the sensors to collect the data. Using NOMA it collects data simultaneously from a near and far sensor paired together. This implies that using NOMA we achieve an optimal trajectory for the UAV ensuring maximum data collection in the limited flight time. Moreover, the trajectories in Figure 3 also illustrate the adaptation of the UAV to changes in various scenario, i.e., for different IoT positions, without requiring retraining.

In this work, the UAV altitude is fixed and the UAV is assumed to hover within each time slot. These assumptions are widely used in the literature to maintain tractability and to focus on the integration of NOMA with trajectory optimization. Nevertheless, optimizing the UAV altitude jointly with its horizontal trajectory, or considering more flexible mobility patterns such as curved or continuous-time trajectories, could further improve system performance. Incorporating such extensions, however, would require a significantly more complex state–action design in the reinforcement learning framework, and is left as a promising direction for future research. Furthermore, for fair comparison, the baseline considered in this study is a TDMA-based UAV trajectory optimization framework implemented under a similar DQN learning setup. This allows consistent evaluation of the effect of incorporating uplink NOMA while maintaining identical environmental and network parameters.

V. CONCLUSION

This paper investigates DQN with NOMA to optimize the trajectory of UAV for maximum data collection from sensors randomly distributed in an open farm field. A novel reward function is developed with appropriate rewards and penalties to achieve optimal path for UAV while collecting data. To collect the maximum possible data from the sensors within the given flying time optimization is carried out using Markov Decision Process. From the achieved results, it can be concluded that using NOMA in place of traditional multiple access techniques like TDMA increases the average data collection percentage from 74.5% to 93.5%, i.e., an increase of 19% was achieved. Furthermore, DQN guaranteed adaptation of UAV to variations in the number of sensors, positions of sensors and flying time of UAV without requiring retraining.

VI. LIMITATIONS AND FUTURE WORK

Although the proposed framework demonstrates significant improvements in UAV-assisted IoT data collection, several limitations remain. This work focuses on a 2D UAV trajectory with fixed altitude, which simplifies analysis but does not capture full 3D mobility; extending the framework to 3D flight is a natural next step. A predominantly LoS channel model is assumed, and future studies may incorporate probabilistic NLoS effects for more realistic environments. The current design employs two-user NOMA pairing per time slot; exploring multi-user clustering and hybrid access schemes remains open. Finally, UAV and sensor energy consumption is not explicitly modeled and can be integrated into future optimization objectives.

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